

Security Monitoring & Incident Response Report

Project: Security Monitoring & Incident Response
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Executive Summary

This project focuses on monitoring Linux authentication and system logs to identify suspicious activities and respond to potential security incidents. Authentication logs were reviewed to distinguish normal user activity from malicious login attempts. The investigation identified repeated failed SSH login attempts consistent with a brute-force attack. Detection logic, incident classification, and security recommendations were documented based on the collected evidence.

Objective

- Analyze Linux authentication logs.
- Identify suspicious login attempts.
- Classify the detected incident.
- Recommend security improvements.

Scope

The project covers analysis of `auth.log` and `syslog.log`, identification of normal login activity, administrative activity, suspicious login attempts, and documentation of the incident response process.

Tools Used

- Visual Studio Code
- `auth.log`
- `syslog.log`
- Markdown Documentation

Methodology

1. Review authentication and system logs.
2. Separate normal and suspicious activities.
3. Correlate IP addresses, usernames and timestamps.
4. Classify the incident.
5. Recommend mitigation measures.

Log Analysis

The authentication logs showed legitimate SSH logins and authorized administrative activities. Repeated failed SSH login attempts from IP address 198.51.100.23 targeted several usernames including admin, postgres, oracle and www-data. These repeated failures indicate automated username enumeration and a brute-force attack. No confirmed successful compromise was observed in the reviewed logs.

Detection Logic

Alerts should be generated when multiple failed SSH login attempts originate from a single IP address, when several invalid usernames are attempted, and when repeated authentication failures occur within a short period.

Incident Classification

Incident Type: SSH Brute-Force Attack

Severity: Critical

Target Service: SSH

Status: Detected and Investigated

Potential Impact: Unauthorized access if valid credentials are compromised.

Security Recommendations

- Enable Multi-Factor Authentication (MFA).
- Use SSH key-based authentication.
- Deploy Fail2Ban.
- Enforce strong password policies.
- Monitor authentication logs continuously.
- Restrict SSH access to trusted networks.

Conclusion

The project successfully demonstrated practical log analysis and incident response techniques. Suspicious SSH authentication attempts were identified, appropriate detection logic was created, and security recommendations were proposed to improve the overall security posture.